Biological Control of Coconut Black-Headed Caterpillar *Opisina arenosella* using *Goniozus nephantidis* and *Cardiastethus exiguus*

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Correct Citation: Venkatesan, T., Ballal, C. R. and Rabindra, R. J. 2008. *Biological Control of Coconut Black-Headed Caterpillar, Opisina arenosella using Goniozus nephantidis and Cardiastethus exiguus*, Technical Bulletin No. 39, Project Directorate of Biological Control, Bangalore – 560 024.

Publisher: Project Director, Project Directorate of Biological Control, Indian Council of Agricultural Research, Bangalore – 560 024.

Printed at: Brilliant Printers Private Limited, #72, Lottegollahalli, RMV 2nd Stage, Bangalore – 560 094.

PREFACE

Coconut is one of the important oilseed crops cultivated in India. Among the different pests that attack coconut, the black-headed caterpillar, *Opisina arenosella* Walker causes severe damage to the palm by defoliation. Use of chemical pesticides is not feasible to control this pest as it leads to destruction of natural enemies, pollinators and honey bees, human/cattle poisoning, bio-magnification and accumulation of poisonous residues in non-target niches, development of pesticide resistance, secondary pest outbreaks, pesticide residues and soil and water pollution. Inundative releases of mass bred natural enemies are known to provide relief against this pest and the biocontrol approach is also eco-friendly, cheap, effective and long lasting. Biological control of the black-headed caterpillar using *Goniozus nephantidis* and *Cardiastethus exiguus* was found to be effective. During the year 2005, about 48,023 coconut palms were affected by *O. arenosella* in Erode district of Tamil Nadu. Nearly 9 lakh *G. nephantidis* were mass produced and released and spectacular control of this pest was achieved. Field trials at Trichur in Kerala indicated the effectiveness of both *G. nephantidis* and *C. exiguus* as potential bio-agents of the black-headed caterpillar.

This bulletin "Biological Control of Coconut Leaf eating Caterpillar, *Opisina arenosella* using *Goniozus nephantidis* and *Cardiastethus exiguus*" provides information on the mass production and utilization of the potential parasitoid *G. nephantidis* and the predator *C. exiguus*. The production costs, storage and packing techniques, dosage, method of release and results of field evaluation studies are detailed in the bulletin. The bulletin would be very useful for all the stakeholders who are involved in biological control attempts against the coconut leaf eating caterpillar.

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INTRODUCTION

The coconut palm *Cocos nucifera* L. is one of the traditional crops grown in Kerala, Karnataka, Tamil Nadu, Andhra Pradesh, Orissa, Gujarat, Maharashtra and West Bengal. The crop provides food, shelter, fuel, medicine and employment to millions of people in the tropics. Recently its cultivation has also been taken up in Madhya Pradesh and the Northeastern states. India is the largest producer of coconuts in the world, producing 12,597 million nuts from an area of 18.40 lakh hectares with a productivity of 6,847 nuts per hectare against a world average of 4,545 coconuts. The crop annually contributes nearly Rs. 7,000 crores to India's GDP and earns a foreign exchange of Rs. 340 crore. In India, ten million people are engaged in cultivation, processing and marketing of coconut.

1. COCONUT BLACK-HEADED CATERPILLAR *OPISINA ARENOSELLA* WALKER (CRYPTOPHASIDAE: LEPIDOPTERA)

Coconut palm is infested by a number of pests in India and the black-headed caterpillar (BHC) *O. arenosella* is one of the most serious pests. It is an endemic pest of coconut in India and Sri Lanka and is also reported from Myanmar and Bangladesh (Cock and Perera, 1987). The pest was first recorded during 1907 from Coimbatore and is now reported in most of the coconut growing areas. In India, this pest occurs in Andhra Pradesh, Gujarat, Karnataka, Kerala, Maharashtra, Orissa and Tamil Nadu. It is an outbreak pest and it assumes severe proportions in the coastal and backwater tracts. It is also observed in the interior areas near river and paddy fields. The pest infests the coconut palm throughout the year at mild to medium intensities, but under

favourable conditions, sporadic outbreak leads to severe depredations to coconut plantations. Peak occurrence of the pest is from February to May in the West coast and April to May in the East coast of India. Besides coconut, the pest also attacks several other host plants such as palmyra palm, *Borassus flabellifer* Linn., date palm, *Phoenix dactylifera* Linn., talipot palm, *Corypha umbraculifera* Linn., wild date palm, *Phoenix sylvestris* Roxb., ornamental palm, *Oreodoxa oleracea* Mart., fan palm, *Livistona chinensis* and banana.

1.1 Biology and Damage Symptoms

Female of *O. arenosella* lays eggs on the lower surface of the leaves with an average fecundity of 200 eggs/female. Egg stage lasts for 5 days. The larval period is 37-42 days, with eight larval instars and pupal period is 12 days. Male longevity is 5 days and female 7 days. The caterpillars of *O. arenosella* live in galleries made of silk and frass on the underside of the mature leaves and feed on the green tissue. Severe infestation often results in a drastic reduction in yield. Massive damage to the crown of leaves affects the thatching quality of leaves, in addition to decline in yield in the years following the outbreak of the pest. In severe outbreaks, thousands of palms are affected and all the fronds, except the 3 or 4 youngest, are destroyed. In the following year, coconut production may be halved due to reduced production of flower spikes, increase in premature nut fall, constriction of the trunk and retardation of growth.

2. BIOLOGICAL SUPPRESSION

2.1 *Goniozus nephantidis* (Muesebeck) (Hymenoptera: Bethylidae)

The coconut ecosystem is rich in predators and parasitoids. In India, 40 species of parasitoids and 50 species of predators have been recorded on *O. arenosella*. Among these, *G. nephantidis* is a dominant parasitoid and is responsible for the reduction of *O. arenosella* population (Dharmaraju, 1963; Cock and Perera, 1987; Venkatesan *et al.*, 2006). This parasitoid is widely used in the biological control programmes against the coconut BHC. It is a sturdy gregarious larval or prepupal ectoparasitoid. It occurs throughout the year, peak activity being observed in February. Its development is positively influenced by warm and dry weather. Generally low parasitism is observed during the humid months. The female practices maternal care of its eggs and the hatching larvae. The parasitoid paralyzes the host larvae and even feeds on the host body fluid. The parasitoid is thus capable of suppressing part of the pest population by merely stinging and paralyzing the 1st - 2nd instar larvae (Sundaramurthy and Santhanakrishnan, 1978, 1979).

The total developmental period is 10-12 days. The egg stage lasts for a day. The larval period is 2-3 days and the pupal period is 7-8 days. Female progeny ranges from 85-90%. Female longevity ranges from 44 to 61 days. Fecundity varies from 65 to 79.5. The adult female is known to have three cycles of oviposition and brood care, each lasting for 12-13 days and separated by 4-5 days. Each female can parasitize 6-8 caterpillars. Biology of *G. nephantidis* was studied on three different host insects (viz., *Corcyra cephalonica*, *Galleria mellonella* and *O. arenosella*). Highest survival was obtained on *C. cephalonica* (88%), followed by *O. arenosella* (86%) and *G. mellonella* (80%) (Venkatesan *et al.*, 2003). Life-table and intrinsic rate of natural increase of *G. nephantidis* was studied on *O. arenosella*, *C. cephalonica* and *G. mellonella*. Highest net reproductive rate was obtained on *C. cephalonica* (42.6 females/female) followed by *O. arenosella* (38.2) and *G. mellonella* (28.8). The intrinsic rate of increase and weekly multiplication rates were also highest on *C. cephalonica* followed by *O. arenosella* and *G. mellonella* (Venkatesan *et al.*, 2004).

Parasitizing Efficiency

The occurrence of *G. nephantidis* throughout the year with higher proportion of female progeny, greater host searching ability, high temperature tolerance and maintenance of density in relation with that of the pest population makes it a versatile parasitoid for the management of *O. arenosella*. In India, different levels of parasitism by *G. nephantidis* have been recorded in various states, viz., 3.7 to 47.6 per cent in Kollam district of Kerala; 57.6 per cent in Thane district of Maharashtra; 51.0 per cent in Mahuva district of Gujarat; 28.0 per cent in Guntur district of Andhra Pradesh and 48.0 per cent in Bangalore in Karnataka.

Production of Laboratory Host, *C. cephalonica*

C. cephalonica is used as a low-cost laboratory host for the mass production of *G. nephantidis*. Therefore, a vigorous and healthy *C. cephalonica* larval colony is a pre-requisite for the continuous production of *G. nephantidis*.

Facilities required:

- Working tables (4.5 × 2.5 ft)
- Slotted angle iron racks (2m × 1m × 45 cm with 6 partitions) for keeping the wooden *Corcyra* boxes
- Wooden boxes for *Corcyra* rearing (45 × 30 × 15 cm)
- Aluminium cups for preventing ants from entering the boxes
- Scissors and brushes
- Refrigerator (150 l capacity) for storing various diet ingredients
- Exhaust fan
- Cotton
- Honey
- Crushed bajra (*Pennisetum glaucum*) grains and groundnut
- *Corcyra* eggs
- Streptomycin sulphate and yeast tablets

Production procedure:

1. Procure bold bajra grains (meant for human consumption) free of pesticide residues. (To test for pesticide residues, a sample containing 100 g from each bag is crushed and 50 first or second instar larvae are allowed to feed for 2-3 days to find out whether the grain has previously been treated with any pesticides and the conclusion could be drawn based on the mortality of the larvae).
2. The required quantity of grain is milled to make 3-4 pieces of each grain and sun-dried.
3. *Corcyra* rearing boxes are thoroughly cleaned and sun-dried before use.
4. In 2.5 kg of grain dispensed in each box, broken groundnut @ 100 g/box and dry yeast tablets (powdered) as a nutritional supplement @ 10 tablets/box are added. Each box is charged with 0.5 cc of *Corcyra* eggs. 0.5 g of streptomycin sulphate is added to control bacterial infections (Jalali *et al*., 2003).
5. After 40 days, the mature *C. cephalonica* larvae are collected from the *Corcyra* boxes for the parasitoid production (Two hundred and twenty five *Corcyra* boxes can be charged for obtaining 1,35,000 *Corcyra* larvae).

Precautions:

It is advisable to observe the following precautions while mass producing *C. cephalonica*:

1. All the windows of the moth production room must be provided with 60 to 100-size brass mesh to prevent the entry of *Bracon hebetor*, a natural enemy of *C. cephalonica*.
2. Date of charging of the grain with *Corcyra* eggs should be recorded on the rearing boxes in order to understand the duration for which each box can be used.
3. When the used contents of the *Corcyra* boxes are discarded, ensure that they are burnt as early as possible to avoid breeding of *B. hebetor* in the open. Alternatively, spent feed can be used for production of *Trichoderma*.
4. The *Corcyra* boxes/basins should be cleaned properly and sun-dried before re-use.
5. Sometimes mites attack the culture in *Corcyra* boxes. Paint a band of sulphur on all the four sides, if mites are noticed.
6. If *B. hebetor* infestation is noticed in the *Corcyra* boxes, all infested boxes should be identified and isolated. Malathion (0.05 %) / endosulfan (0.07%) can be sprayed on the outer surface of all the boxes. This treatment does not kill *Corcyra*.
7. For controlling the predatory mite, *Pyemotes ventricosus*, a thin layer of wettable sulphur can be applied on the surface of the bajra grains inside the box.

Mass Production of *G. nephantidis*

Facilities required (Details given in Annexure):

- Glass tubes (9 × 2.5 cm)

- Plastic trays (2 × 2 feet) for keeping the glass tubes
- Pearpet containers (1 lit. cap.) for rearing the adult parasitoids
- Plastic boxes (10 × 5 cm) for keeping the parasitoid cocoons
- Ceiling fan
- Exhaust fan
- Working tables (4.5 × 2.5 ft) and stools

Production procedure:

Stock culture of *G. nephantidis* is collected from *Opisina*-infested gardens and maintained on the mature larvae of *C. cephalonica* in the laboratory. Males and females of *G. nephantidis* are released into a pearlpet container of 1 lit capacity for mating. Males and females of *G. nephantidis* can be easily differentiated: Females are larger in size with prominent ovipositor and males are smaller with a blunt abdominal tip. Cotton swabs moistened with dilute honey (50%) and water are provided as adult food. After a preoviposition period of 6 days, the females are separated and kept in glass vials individually, as there is contest behaviour among females which involves aggressive postures, rapid chases and violent fights in which contestants attempt to bite and sting each other. Mature and healthy *Corcyra* larvae (5th - 7th instar) are released individually into each vial. If small larvae are used, it leads to weak individuals of *G. nephantidis*. The stock culture of *G. nephantidis* can be reared using the natural target host *O. arenosella* once in 6 generations to maintain the vigour of the parasitoid.

The parasitoid stings and paralyzes the host larvae and lay their eggs outside the host body. The females lay a clutch 8-18 eggs per host larva, approximately a day after paralyzing the host. Following oviposition, the female parasitoid remains with its broods until the progeny larvae reach an advanced stage of development. Parasitized larvae are removed daily using fine forceps (by locating the eggs of *G. nephantidis* on the body of larvae) and placed on a tissue paper kept in a ventilated plastic box (10 × 5 cm). The larvae of *G. nephantidis* feed on the host larvae resulting in shriveling and death of the host larvae. The parasitoids when fully developed spin white/pale white cocoons adjacent to the host. Cocoon formation occurs in 24 hours. The parasitoid completes its life cycle in 12 days. A temperature of 27 ± 1 °C and RH of 60 ± 5% should be maintained in the production units.

Prior to mass production of the parasitoids, the stock culture should be increased to 5000 consisting of males and females. About 600 healthy mature *C. cephalonica* larvae can be obtained from each *Corcyra* box. 45,000 mature *Corcyra* larvae could be obtained from 75 *Corcyra* boxes, at 45 days after charging @ 1000 larvae/day and these can be used for *G. nephantidis* production. Similarly, another 45,000 larvae each could be obtained from second and third batches. An average of six cocoons can be obtained per parasitized larva. About 2,500 *Corcyra* larvae can be exposed to the parasitoids and 30,000 cocoons can be produced in a week in two exposures. Among these, 5,000 parasitoids are to be kept for the next exposure and the remaining are sent for field releases. By following this method, required number of *Goniozus* can be produced in a period of 4 months. Certain quality parameters of the parasitoid should be taken into consideration during the production (Table 1).

Table 1: Quality parameters of *G. nephantidis* to be considered during production

Parameters	Standard (≥)
Female Weight (100 nos.)	14.5 mg
Per cent survival	87.0
Longevity (days)	58.0
Fecundity (nos./female)	48.0
Percentage female progeny	85.0
Parasitizing efficiency (no. of larvae/female)	8.0

Cost of Mass Production of *G. nephantidis*

The cost of production is calculated on the basis of the production of the host insect as well as the parasitoid. The cost of production of 1,00,000 *G. nephantidis* cocoons is Rs. 27,600/-. Cost-Benefit Ratio (CBR) in the production of parasitoid is 1:1.45 (Economics for the production of 3.25 lakhs *G. nephantidis* cocoons is given in Annexure). The technology is highly

feasible and can be adopted for the large scale production of the parasitoid. Localized production of *G. nephantidis* could be taken up at village or district level especially by farmers, unemployed graduates, private and public sector units and NGOs. This module can be scaled up to any level (Venkatesan *et al.*, 2007).

Storage, Packing and Shipment of *G. nephantidis* Cocoons

Storage:

For evaluating the efficacy of *G. nephantidis* against *O. arenosella*, it is imperative to ensure that a large number of parasitoids are available at the appropriate time for their mass release. The optimum low temperature and duration for storage of 48-h-old *G. nephantidis* cocoons was found to be 15 °C and 10 days, respectively. It resulted in 90% adult emergence and the emerging females had a fecundity of 84 eggs/female and a longevity of 57 days. These parameters were comparable to those of the adults emerging from cocoons kept at ambient temperature (26 ± 1 °C) (Venkatesan *et al.*, 2000). This technique would be useful for continuous laboratory rearing and for planning timely field releases.

Packing and shipment:

Tissue paper pieces containing *Goniozus* cocoons are gently removed and kept in polythene covers (200 nos/cover) and packed neatly. Adequate care is taken to prevent damage to the cocoons. Pinholes are provided for ventilation and honey streaks for adult feeding as adults may emerge in transit. These cocoons in the polythene pouches are packed in plastic boxes (10 × 5 cm) and shipped through courier or speed post services.

Field Evaluation of *G. nephantidis*

Augmentative releases of laboratory cultured parasitoids have played a very substantial role in attempts to control *O. arenosella* in both India and Sri Lanka. Successful biological control of the pest is possible only if large scale field releases of the parasitoids and/or predators are made in a wider area.

Method of Field Release of *G. nephantidis*

The method of release of *G. nephantidis* adopted earlier was climbing the palm and releasing in the crown region. Besides being expensive, it is very difficult to get climbers. A devise mounted on a long pole to release the parasitoids at the crown region has also been developed but not used widely. Hence, the aim was to evolve a farmer-friendly, easy and economical method of release of *G. nephantidis*. Repeated trials revealed that 90-100 per cent of the parasitoids released at the trunk (1.2 m from the ground level) reached the crown of the palm, irrespective of the height of the tree. The mean time taken by each batch of the parasitoids to reach the crown varied from 13 to 22 minutes. It was also found that the parasitoids preferred to crawl on the tree trunk to reach the crown rather than fly. Based on this observation, it is suggested that *G. nephantidis* adults could be released at the trunk (at 1.2 m height from the ground level) of the coconut palm for the management of *O. arenosella* instead of releasing at the crown region of the palm or arbitrarily on unit area basis. Lyla *et al.* (2006a) evaluated and compared two methods of release i.e., trunk and crown release and reported that there was no significant difference between the two methods of release of *G. nephantidis* in reducing the population of *O. arenosella* (Table 2).

Table 2: Reduction in *O. arenosella* population after release of *G. nephantidis* (Lyla *et al.*, 2006a)

Treatment	Percentage reduction in *Opisina* population after release		
	I	III	VI
Release at 1.2-1.5m height of palm	73.7[b]	82.0[b]	96.0[b]
Release at crown	62.2[b]	56.7[b]	90.0[b]
Control	62.5[a]	77.8[a]	-50.5[a]

Figures in a column followed by the different alphabets are significantly different (P ≤ 0.05).

A field release experiment in Channapatna has shown that four releases of the parasitoid @ 10/palm gave significant control of the pest population (Venkatesan *et al.*, 2006).

Cost of Field Release

Cost of release of *G. nephantidis* @ 10 cocoons per palm was computed based on the commercial cost of *Goniozus* cocoons (@ Rs. 0.30/cocoon). With the given spacing of 7.5 metres between trees and between rows, followed by coconut farmers in India, the total number of palms/ha would be 175. The cost of release of the parasitoid @ 10 per palm per ha for four releases would be Rs. 2100/-. Increase in the dosage of the parasitoids from 10 to 30 per palm led to an increase in cost of release, but did not contribute to any significant decrease in pest population.

2.2 Anthocorid Predator – *Cardiastethus exiguus* Poppius

Cardiastethus exiguus Poppius (= *C. pygmaeus pauliani*) was initially reported from nests of *Ploceus philippinus* in south India by Muraleedharan (1975). Abdurahiman *et al.* (1982) reported for the first time the presence of *C. exiguus* in Kerala as a predator of eggs and young larvae of *O. arenosella*. Earlier reports have indicated that the population of this predator fluctuates with that of *O. arenosella* indicating the correlation of this predator with the pest.

Biology

The eggs of *C. exiguus* are creamy white, cylindrical, slightly curved with an operculum. The nymph is reddish with three dark spots on the 3rd, 4th and 5th abdominal segments. Newly emerged adults are pale red with creamy white wings and gradually take on a dark brown colour. When fed on the natural host, the ovipositing female can live for 45-50 days and virgin females for 120 days. The fecundity is 75-200 eggs. Earlier studies have revealed that this predator could feed on 200-225 eggs of *O. arenosella* during its life time and that oviposition stopped when food (*Opisina* eggs) supply stopped.

When reared on *C. cephalonica* eggs, the incubation period of *C. exiguus* is 3-5 days and the nymphal period 14-21 days. Mean male longevity is 35.75 days and female longevity 66.07 days. Egg laying can be obtained on cotton in the laboratory and the fecundity ranges between 21 and 64 eggs per female. Amongst the eggs laid, 65-85% hatched successfully and about 40% of the progeny are females.

Sexual Dimorphism

The adult male and female can be differentiated based on the following characters: The posterior margin of the last four abdominal sternites of the male is skewed towards the lateral side and the anterior margin of the last sternite is laterally excised from the penultimate sternite. In the female the posterior margin of the seventh sternite has a V-shaped median projection bisecting the eighth sternite. The antennae are also sexually dimorphic: segment II is distinctly stout and sausage shaped in the male, but thinner and gradually enlarged towards apex only in the distal half in the female. Segments III and IV are also thick and spindle shaped in the male, but relatively more slender in the female.

Feeding Potential

Cardiastethus exiguus exhibits cannibalism when food is scarce and so it is very important to provide adequate food while mass rearing this anthocorid. During the nymphal period, when fed on *C. cephalonica* eggs, the total feeding is about 31-47 eggs and feeding per day ranges between 1 and 4 eggs. Adult males can feed on a total of 208.25 eggs and mean feeding per day is 5.14 eggs. Adult females feed on more number of eggs in comparison to males. Throughout its longevity, one adult female can feed on a total of 373.75 eggs and 6.23 eggs per day.

Day-wise Feeding Potential

Peaks of feeding are observed on 6th, 10th, 13th and 14th days from the date of hatching for nymphs. In the case of adult males, some peaks showing higher feeding rate are evident on days 2, 6, 14, 20, 22, 30 and 33, and during these days, feeding per day is 7 to 8 eggs. In the case of adult females, it is observed that feeding ranges between 3 and 8 eggs till 23rd day. After this, there are some higher peaks of feeding on days 24, 28 to 30, 32, 34 and 40 during which period 9 to 11 eggs are fed on per day (which is higher than the male feeding potential). From the 41st day, feeding rate is 6 or less per day. The feeding potential of adult females during the last phase of its longevity is similar to that of males.

Based on the feeding potential studies, we can decide on the number of *Corcyra* eggs to be provided to the nymphs and adults of different ages. It is necessary to provide adequate number of eggs for the predator to avoid cannibalism. The nymphs can be provided with 4 eggs per day and the adults with 10 eggs per day. If there is a shortage of host material, it is possible to provide host eggs in a judicious manner for the nymphs: 3 eggs per day per nymph can be provided for the first five days and 4 eggs per

day till adult formation. For adults, provision of 8 eggs per day for the initial 20 days of their longevity and 10 eggs per day for the next 20 days. If the adults survive beyond 60 days, 6 eggs per day per adult can be provided. It is found that for the production of 450 *C. exiguus* adults, 1 cc of *C. cephalonica* eggs is required. This information generated in this study can be used for planning the mass production of this predator utilizing *C. cephalonica* eggs.

Mass Production Method of *C. exiguus*

At the Project Directorate of Biological Control the first attempts at mass production and release of this predator were made. The culture of *C. exiguus* was originally obtained from *O. arenosella* infested coconut plantations in Mandya and Mysore areas. A simple mass production method has been devised based on the information derived from the biology and feeding potential studies. *C. exiguus* adults (30 males and 30 females) are placed in pearlpet containers (500 ml capacity) provided with cotton strands, UV-irradiated *Corcyra* eggs (@ 300 eggs per day), pollen dust and a swab of cotton soaked in water (stuck to the wall of the container). Cotton strands are found to be ideal oviposition substrates. Egg laying is also observed on the walls of the plastic containers.

After 24 h, the eggs are removed and placed in fresh containers for nymphal hatching. Alternatively, the adults can be collected from adult containers and released into a fresh container, allowing the eggs present in the original container to hatch. This method is followed especially when more number of eggs are laid on the container walls, in order to avoid damaging the eggs while attempting to remove them. The container with adults is provided with fresh cotton strands and *Corcyra* eggs at 24 h intervals. After 3-4 days, when nymphs hatch, *Corcyra* eggs are provided @ 4 eggs per day per nymph. In order to avoid cannibalism, care has to be taken to separate *C. exiguus* eggs from their adult stage at 24 h intervals and not to mix nymphs of different ages in the same container. From one container, one can obtain 3000 adults in a span of a month. It is possible to plan the production based on the number required for field releases, by setting up the required number of adult containers.

This anthocorid can thus be multiplied continuously on *C. cephalonica* eggs. However, after continuously multiplying for a year, deterioration in the culture is observed. At this juncture, field collected nymphs or adults have to be brought in for rejuvenating the lab culture.

Cost of Production

By maintaining 150 rearing containers with 30 pairs of *C. exiguus* per container, approximately 6 lakh nymphs can be produced per month. Taking into consideration only the cost of *Corcyra* eggs, the cost of 100 nymphs is estimated at Rs. 1.60 and Rs. 5.80 if cost of equipments and facilities is included. As *Corcyra* eggs are used as common laboratory hosts for both *Goniozus* and *Cardiastethus*, the host production facilities can be shared/production of both the bioagents.

Field Evaluation of *C. exiguus*

At Channapatna in Karnataka, a field trial was conducted to evaluate the performance of *C. exiguus* on *O. arenosella*. Monthly observations were recorded on pest population and *Cardiastethus* population per leaflet. Six field releases were made @ 50 *Cardiastethus* adults/nymphs per tree. The first three releases were made at weekly intervals when young larvae were observed in the samples. Last three releases were made at monthly intervals.

The larval population per leaflet was 0.55 in the pre-treatment observation and during observation V, it was 0.08. The count of anthocorids/leaflet was 0.22 in the pre-treatment count, while during the second post-treatment observation it was 0.68/leaflet. During the last two observations, *O. arenosella* larvae were in late instar and the anthocorid count was also less. This indicates that the anthocorid population was more when the pest population was in the younger stages and hence the releases should concentrate on those stages.

Field releases of *C. exiguus* were carried out against *O. arenosella* during summer months in Kerala. The anthocorid proved to be a very efficient predator and the pest population could be suppressed when three releases of 50 nymphs/adults were made at five day intervals (Lyla *et al.*, 2006). The releases should however coincide with the egg laying stage of the pest (Table 3).

Table 3: Efficacy of *C. exiguus* against *O. arenosella* (Lyla *et al.*, 2006)

Treatment	Mean no. of *Opisina* larvae/palm	
	Pre-count	Post-count after 3 releases
50 *C. exiguus* nymphs or adults/palm	4.65	0.35[b]
100 *C. exiguus* nymphs or adults/palm	4.90	0.25[b]
Control	4.60	2.95[a]

Figures in a column followed by different alphabets are significantly different (P ≤ 0.05).

3. INTEGRATED USE OF *G. NEPHANTIDIS* AND *C. EXIGUUS*

In a large scale validation trial in Kerala, among different treatments, sequential release of *C. exiguus* and *G. nephantidis* either alone or in combination was found to be more effective in checking the BHC population as compared to releases of *Trichogramma embryophagum* adults and untreated control (Table 4).

Table 4: Large scale validation on biological control of *O. arenosella* using different bio-agents in Kerala*

Treatments (Release of)	Mean no. of *Opisina* larvae/leaflet	
	Pre-count	Post-count after 3 releases
Trichogramma embryophagum adults @ 1000/palm	0.4	0.11[bc]
C. exiguus nymphs @ 50/palm	0.5	0.06[bc]
G. nephantidis adults @ 10/palm	0.2	0.10[bc]
Sequential release of *T. embryophagum* & *G. nephantidis*	0.3	0.06[bc]
Sequential release of *C. exiguus* & *G. nephantidis*	0.3	0.01[c]
Untreated control	0.7	0.51[a]

Figures in a column followed by different alphabets are significantly different (P ≤ 0.05).
* Source: Reevi, P. and Lyla, K. R. (KAU, AICRP on Biological Control of Crop Pests and Weeds, Annual Report, PDBC (2008)).

4. CONCLUSION

Augmentative releases of both *G. nephantidis* and *C. exiguus* were found effective in checking the population of coconut black-headed caterpillar *O. arenosella*. It is necessary to train the farmers/extension workers on mass production and field release techniques of these natural enemies. It is also possible to establish production units of the above two bio-agents at village level for the successful control of the pest. Large-scale demonstrations have to be taken up under public-private-people partnership mode to bring greater visibility to the biological control of *O. arenosella*.

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ANNEXURE

Cost of production of 3.25 lakhs *G. nephantidis*

I. Fixed Costs

Sl. No.	Materials	Unit	Quantity	Unit Price (Rs.)	Total Cost (Rs.)	Depreciation @ 20%
1	Wooden boxes	Nos.	225	500	112500.00	22500.00
2	Glass tubes	Pieces	2500	10	25000.00	5000.00
3	Plastic boxes	Pieces	300	8	2400.00	480.00
4	Plastic containers	Pieces	15	30	450.00	90.00
5	Plastic trays	Pieces	20	30	600.00	120.00
6	Steel racks	Nos.	15	2,000	30000.00	6000.00
7	Refrigerator	Nos.	1	10,000	10000.00	2000.00
8	Exhaust fans	Nos.	2	1,500	3000.00	600.00
9	Ceiling Fans	Nos.	2	1,500	3000.00	600.00
Sub total		**—**	**—**	**—**	**186950.00**	**37390.00 (5 yrs) 7478.00 (1 yr)**
Interest on fixed capital (@ 9%)					673.02	8151.02
Half yearly charge					**—**	**4075.51**

II. Variable Costs (Half year)

Sl. No.	Particulars	Unit	Quantity / Period	Unit price (Rs.)	Total Cost (Rs.)
A. Material input costs					
1	Bajra (+ milling & transporting charges)	Qtls.	5.6	800	4480.00
2	Groundnut (+ milling & transporting charges)	Qtls.	0.23	4000	920.00
3	Streptomycin sulphate	Gm.	113	12	1356.00
4	Yeast tablets	Gm.	675	2	1350.00
5	Electricity & water charges	Months	6	500	3000.00
6	Labour charges for *Corcyra* rearing for 6 months	Months	2 nos.	2,500	30000.00
7	Labour charges for parasitoid rearing for 4 months	Months	2 nos.	2,500	20000.00
8	Misc. items (scissors, forceps, tissue rolls, towels, cotton)	—	—	—	4000.00
9	Interest on working capital (9%)	—	—	—	2565.00
Total variable cost (TV)					**67671.00**

III. Total Cost (Half year)

Particulars	Amount (Rs.)
Rent on land for 6 months @ Rs. 3000/- per month (Fixed cost)	18000.00
Depreciation (on fixed cost)	4075.51
Total fixed cost (TF)	**22075.51**
Total variable cost (TV)	67671.00
Total cost for half year (TV + TF)	**89746.51**
Gross Returns (GR)*	**130000.00**
Net Returns [GR - (TV + TF)]	**40253.49**
Cost : Benefit Ratio	**1:1.45**

* Gross returns calculated based on the sale of 3.25 lakh *G. nephantidis* @ Re 0.40/cocoon.